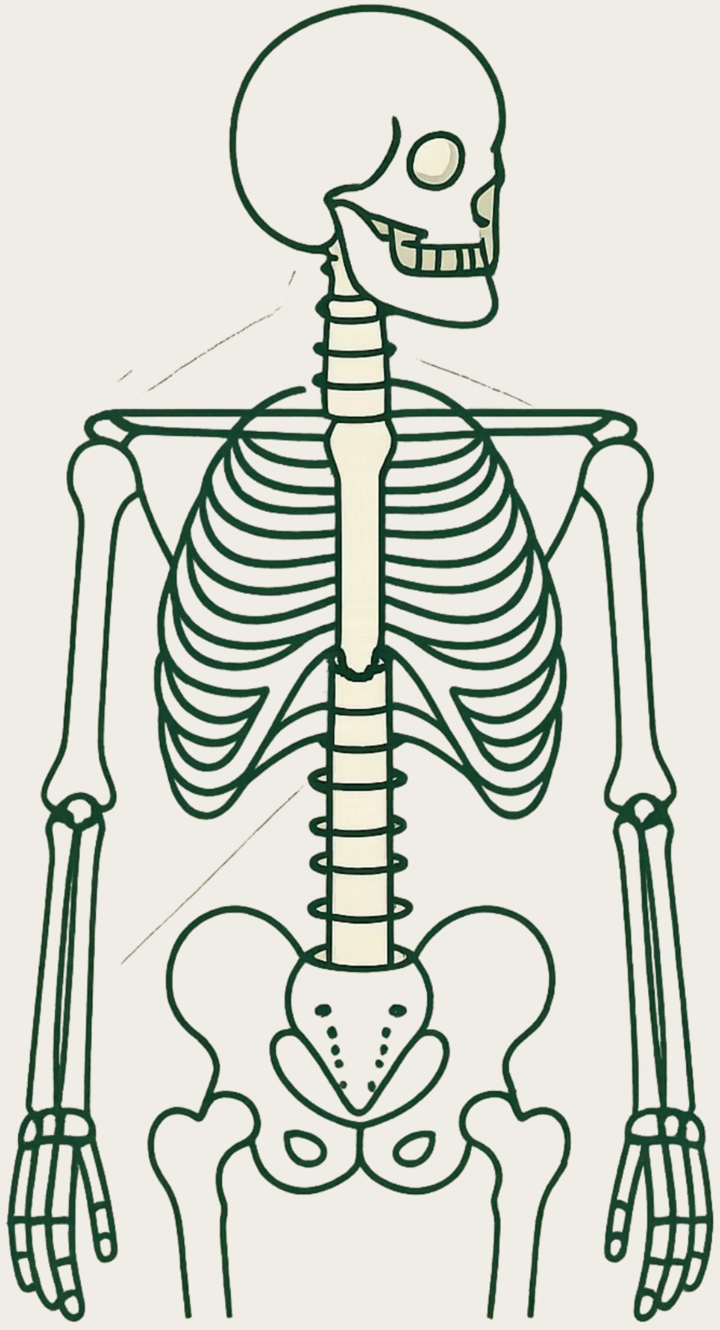
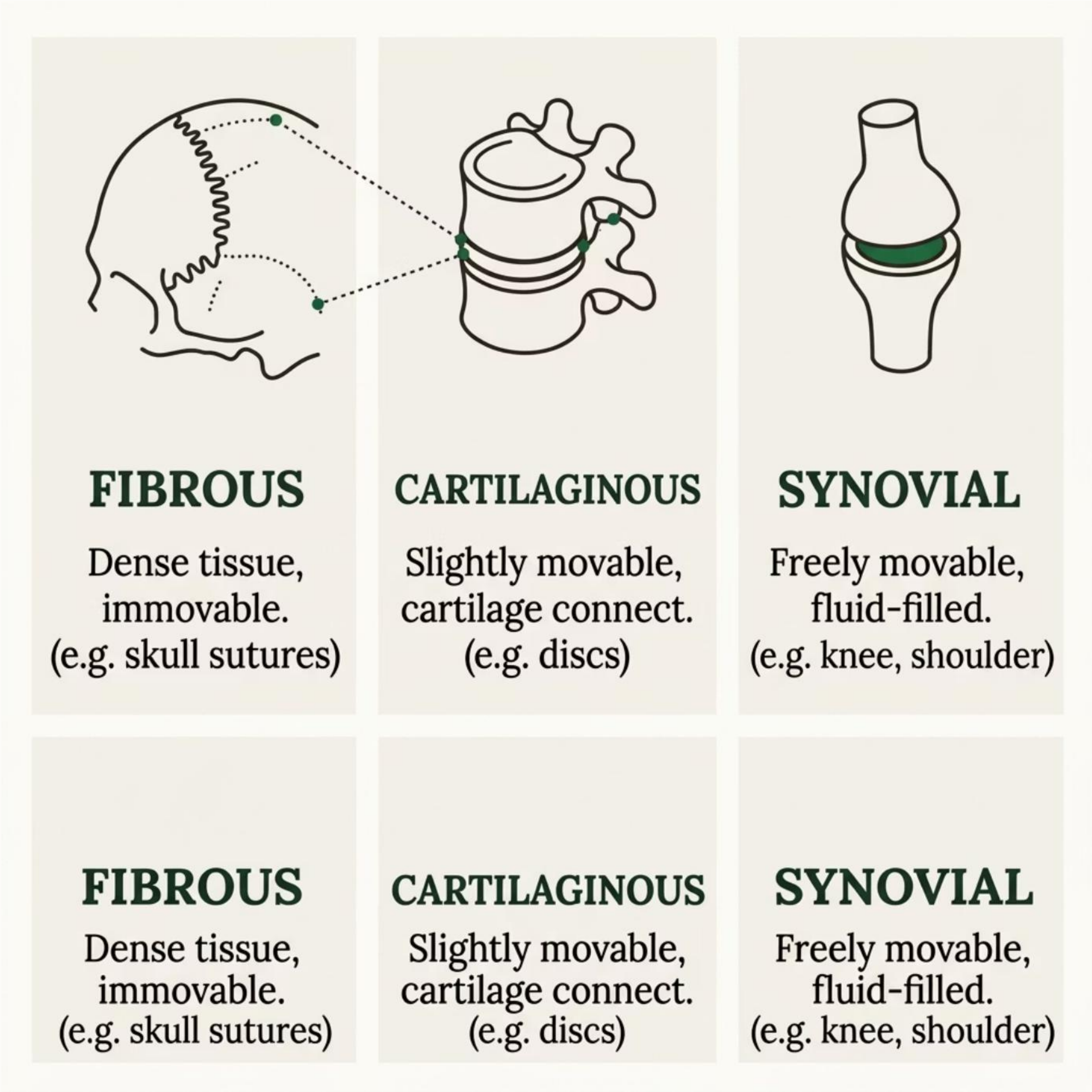
Unit 4: Mechanics of Joints

A comprehensive study of skeletal joint architecture, force mechanics, and Mechanics of shoulder, spinal column, hip, knee and ankle

BIOMECHANICS | HUMAN ANATOMY | JOINT MECHANICS



Architecture of Skeletal Joints

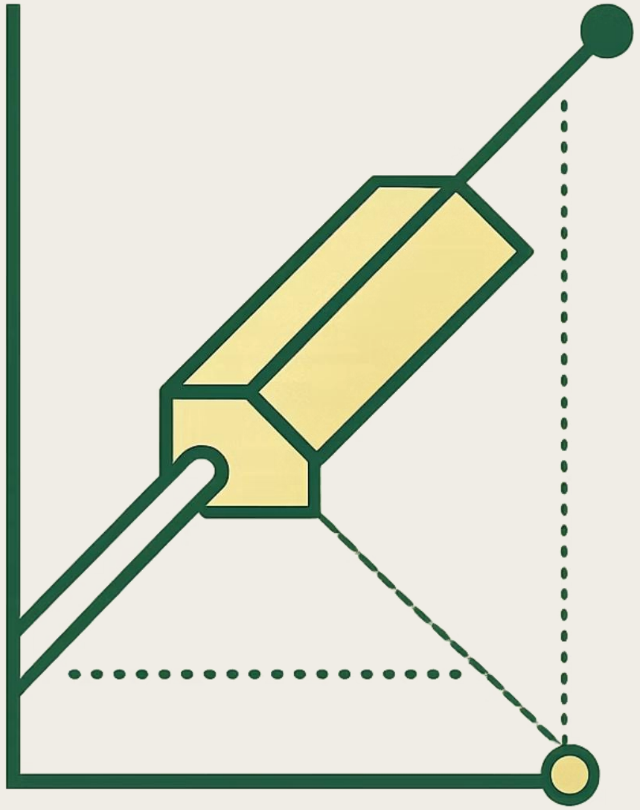


Skeletal joints are classified by structure and mobility. **Synovial joints** — the most common — feature a fluid-filled cavity that enables smooth, low-friction movement.

Articular Cartilage
Reduces friction and absorbs shock

Synovial Fluid
Lubricates and nourishes joint surfaces

Joint Capsule
Encloses and stabilizes the joint



Forces & Stresses in Human Joints

Compressive Force

Pushes joint surfaces together; dominant in weight-bearing joints like the knee and hip during stance phase.

Tensile Force

Pulls structures apart; experienced by ligaments and tendons during joint loading and muscle contraction.

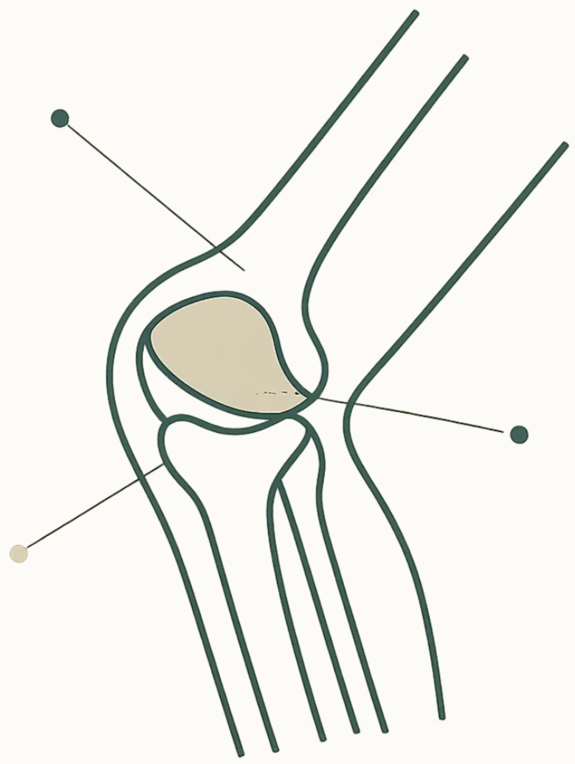
Shear Force

Acts parallel to joint surfaces; a primary cause of cartilage wear and ligament injury.

Torsional Stress

Rotational loading that can damage articular surfaces, especially during pivoting movements.

Mechanics of the Elbow



The elbow is a **hinge joint** formed by three bones: the humerus, ulna, and radius. It allows flexion–extension and forearm rotation (pronation–supination).

→ Flexion: 0°–145°

Driven by biceps brachii and brachialis

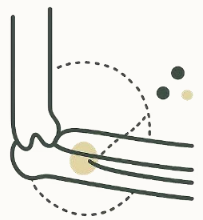
→ Extension: 145°–0°

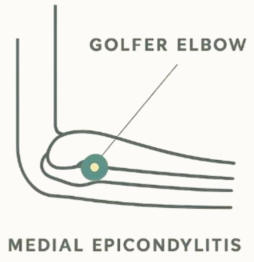
Controlled by triceps brachii

→ Rotation: ~180°

Supination/pronation via radioulnar joint

Common Elbow Injuries





Lateral Epicondylitis

"Tennis elbow" — overuse injury of extensor tendons at the lateral epicondyle. Common in repetitive gripping activities.

Medial Epicondylitis

"Golfer's elbow" — inflammation of flexor-pronator tendons on the inner elbow, caused by repetitive wrist flexion.

Elbow Dislocation

Second most common joint dislocation. Often involves ligament rupture and requires prompt reduction and rehabilitation.

Mechanics of the Shoulder

The glenohumeral joint is the **most mobile joint in the human body** — a ball-and-socket design that trades stability for range of motion.

Flexion / Extension

0°–180° forward; 0°–60° backward

Abduction / Adduction

0°–180° away from body; return to neutral

Internal / External Rotation

~90° each direction about the humeral axis

Circumduction

Combined conical movement through all planes

The Lower Extremity Kinetic Chain

Hip, Knee, and Ankle Mechanics — a technical exploration of how the three major lower limb joints work together to produce movement, absorb load, and transmit force through the body.

BIOMECHANICS | SPORTS SCIENCE | CLINICAL ANATOMY

The Hip: A Master of Motion and Stability

The hip is a **ball-and-socket synovial joint** — the acetabulum cradles the femoral head, providing inherent stability while enabling a wide range of motion across three planes.

Sagittal Plane

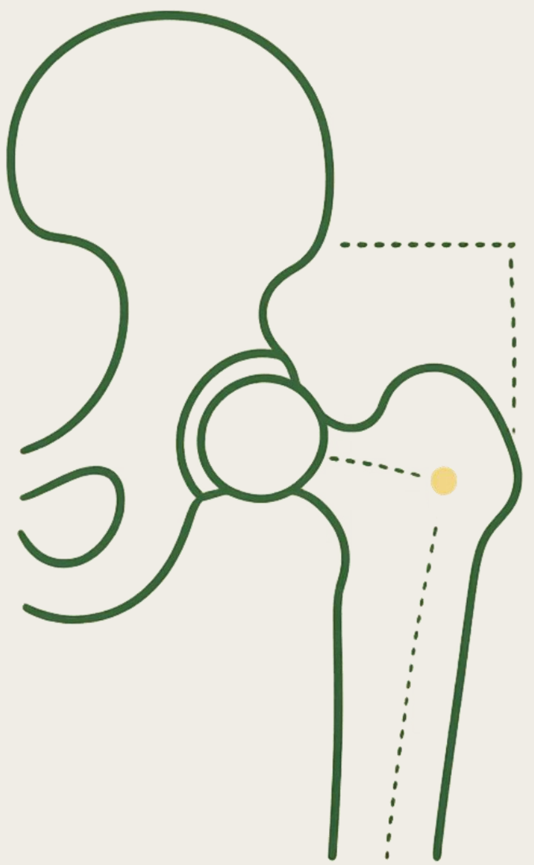
Flexion & Extension

Frontal Plane

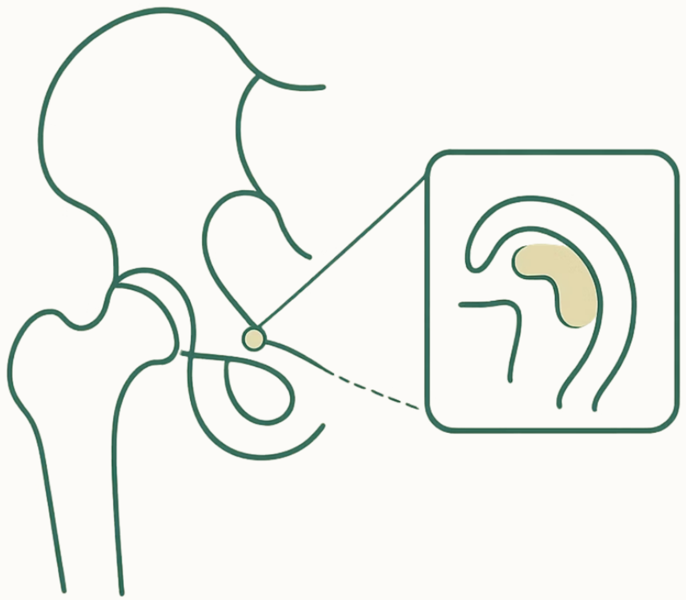
Abduction & Adduction

Transverse Plane

Internal & External Rotation



Hip Injuries: When Movement Becomes Painful



Labral Tears

Damage to the cartilage ring lining the acetabulum, caused by acute trauma or chronic degeneration. Presents with pain and a catching sensation during rotation.

Femoroacetabular Impingement (FAI)

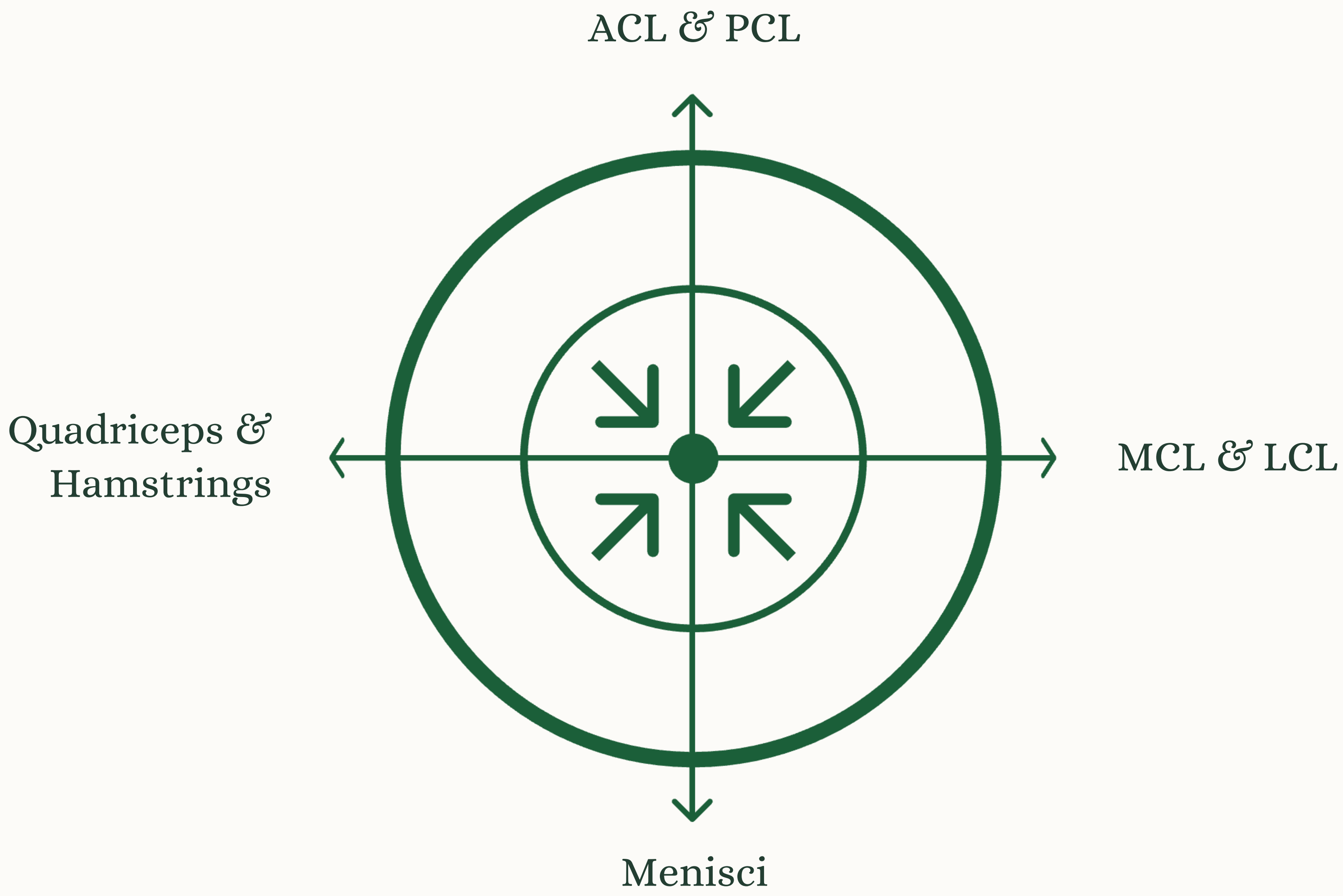
Abnormal bone growth on the femoral head or acetabulum creates friction during movement, leading to progressive joint damage.

Bursitis

Inflammation of the fluid-filled bursae cushioning the joint, typically at the lateral hip. Causes localized pain and tenderness.

The Knee: A Complex Hinge

Far from a simple hinge, the knee permits **six degrees of freedom** — three translational and three rotational movements — stabilized by an intricate network of ligaments, menisci, and dynamic musculature.



The **patella** acts as a biological lever, increasing the mechanical advantage of the quadriceps by altering the tendon's angle of pull on the tibia.

Knee Mechanics & Injury Vulnerabilities

Dynamic Stability

Ligaments and muscles work in concert during walking, running, and cutting. The quadriceps and hamstrings provide active stabilization throughout the gait cycle.

Rotational Stress

Pivoting during leg kicks optimizes power but increases rotational stress on the knee, elevating the risk of **ACL injury** — particularly in non-contact scenarios.

Kinetic Chain Effect

Chronic ankle instability alters knee kinematics during landing, potentially increasing ACL loading — a critical interdependency often overlooked.

The Ankle: Foundation for Movement

Why the Ankle Matters

The ankle complex is the body's primary interface with the ground — responsible for **shock absorption**, terrain adaptation, and propulsion. Dysfunction here cascades upward through the kinetic chain.

> Limited ankle dorsiflexion forces compensatory strategies at the knee, potentially inducing valgus stress or unwanted rotation during loading.

Ankle Structure & Movement

Talocrural Joint

The "true" ankle — articulation of tibia, fibula, and talus. Governs **dorsiflexion** (toes up) and **plantarflexion** (toes down).

Subtalar Joint

Located below the talus. Enables **inversion** (sole tilts inward) and **eversion** (sole tilts outward), critical for adapting to uneven surfaces.

Second-Class Lever

During gait and calf raises, the gastrocnemius and soleus apply force to the heel, using the ball of the foot as a fulcrum to lift body weight.

Ankle Instability & Its Ripple Effect

1 **Chronic Ankle Instability (CAI)**

Alters knee kinematics during landing, mimicking biomechanical patterns associated with ACL injury mechanisms. Reduced knee flexion at peak anterior tibial shear force has been observed in individuals with CAI.

2 **Inversion Sprains**

The most common ankle injury — the foot rolls inward, stressing or tearing the lateral ligaments, particularly the **ATFL**.

3 **Achilles Tendonitis**

Overuse of the body's thickest tendon, connecting the calf muscles to the calcaneus. Common in athletes with high propulsion demands.

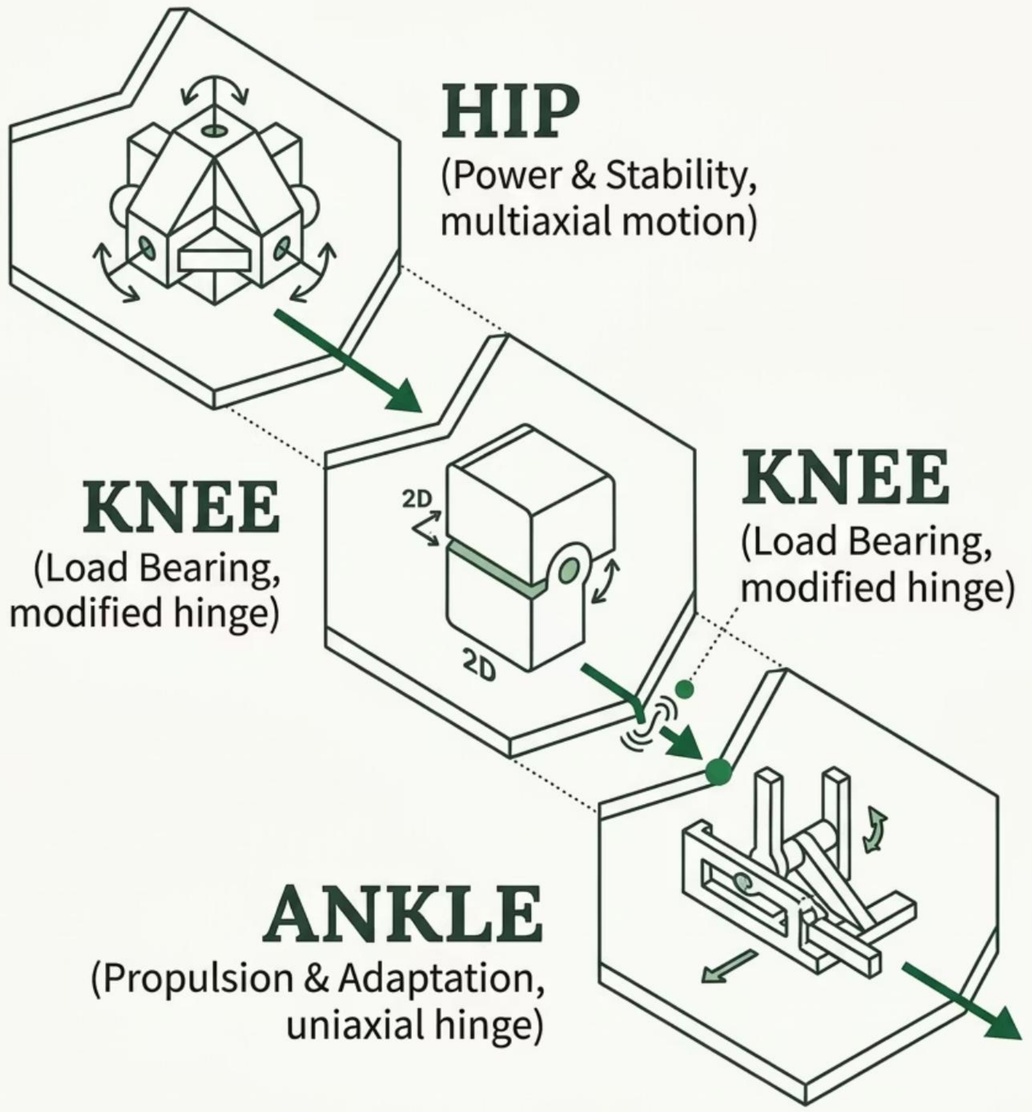
ANKLE SPRAIN

LATERAL LIGAMENTS

The Interconnected Kinetic Chain

The lower extremities function as a **unified kinetic chain** — dysfunction in one joint creates compensatory patterns that propagate to others.

- Altered hip rotation during cutting increases non-contact ACL risk
- Foot strike patterns influence knee and hip loading
- Ankle dorsiflexion limitations force compensatory knee valgus
- Chronic ankle instability changes landing mechanics at the knee



Mastering Lower Extremity Health

Understanding the mechanics, injury patterns, and interdependencies of the hip, knee, and ankle empowers athletes, clinicians, and individuals to make informed decisions for **long-term joint health and peak performance**.

Injury Prevention

Address mobility deficits and movement asymmetries before they become injuries.

Targeted Rehab

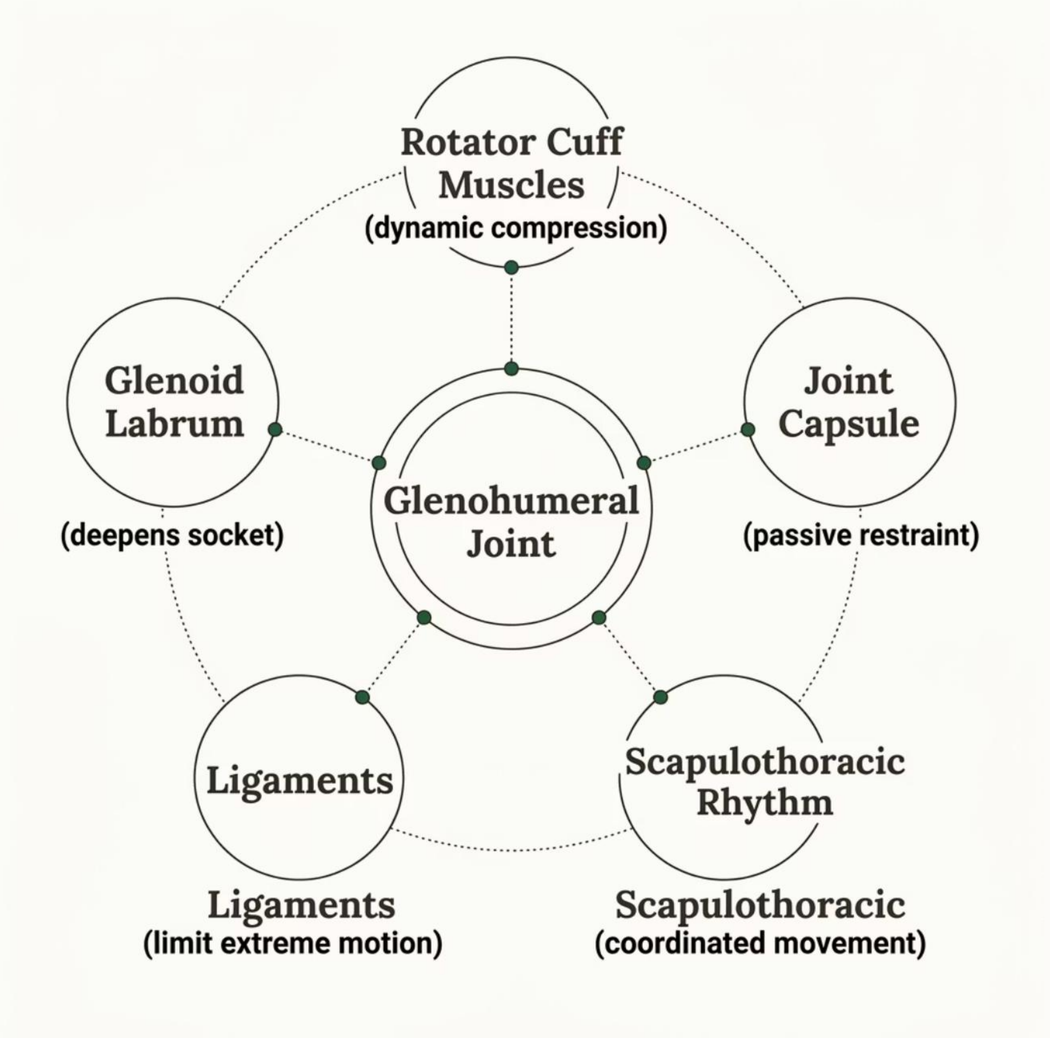
Treat the whole kinetic chain, not just the symptomatic joint.

Performance Optimization

Efficient force transfer through all three joints maximizes athletic output.

Shoulder Stability & the Rotator Cuff

The shallow glenoid fossa provides minimal bony constraint. Stability relies on **dynamic soft-tissue structures**, primarily the rotator cuff muscles and the glenoid labrum.



1

Supraspinatus

Initiates abduction

2

Infraspinatus

External rotation

3

Teres Minor

External rotation

4

Subscapularis

Internal rotation

Mechanics of the Spinal Column

The vertebral column serves as the body's central axis — providing structural support, protecting the spinal cord, and enabling complex multi-planar movement.

1

Cervical (C1–C7)

Most mobile region; supports the head. Allows flexion, extension, rotation, and lateral bending.

2

Thoracic (T1–T12)

Attached to ribs; limited motion. Primary role is protection and posture.

3

Lumbar (L1–L5)

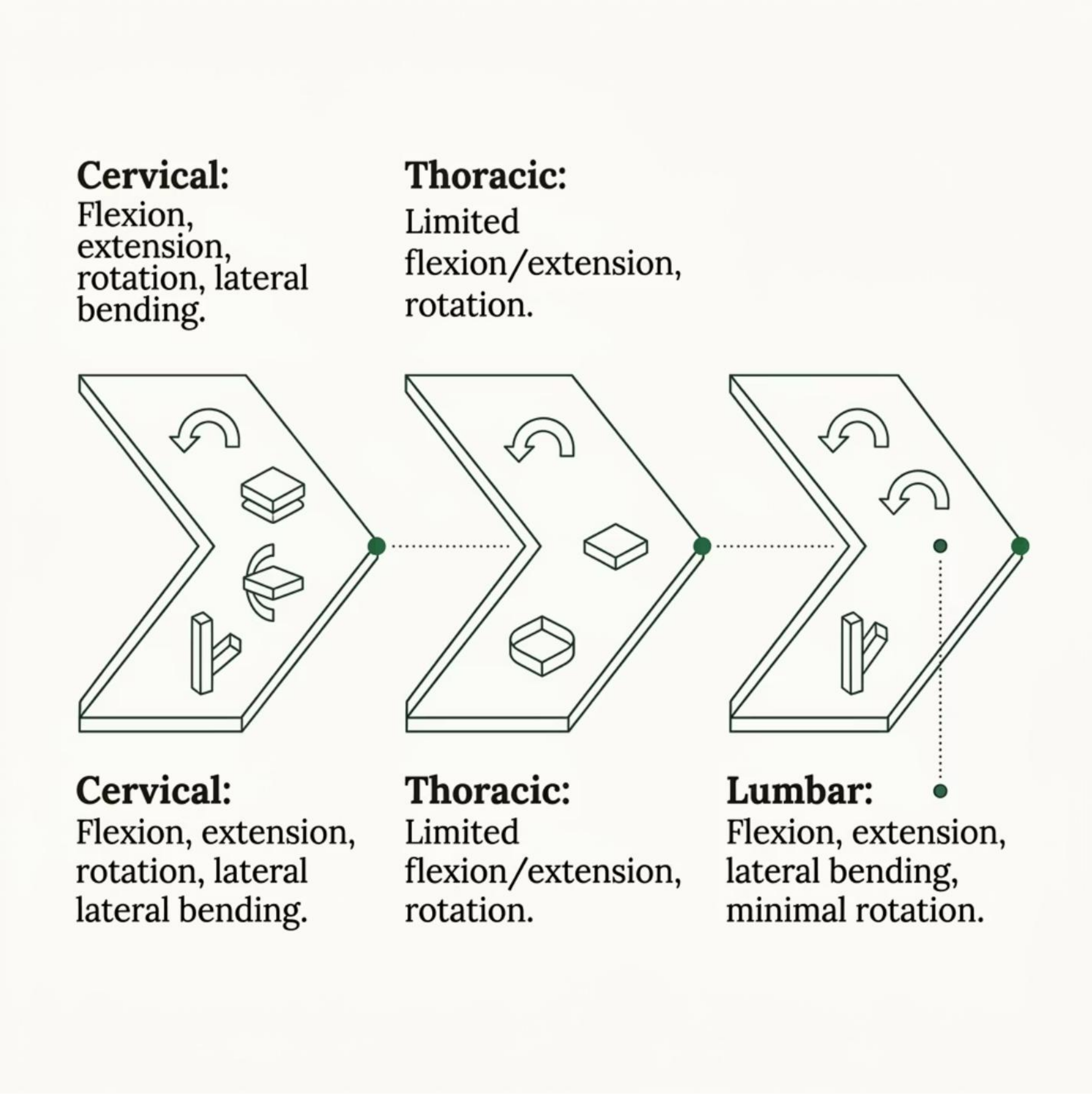
Largest vertebrae; bears greatest load. Allows flexion and extension, limited rotation.

4

Sacrum & Coccyx

Fused segments; transfer load to the pelvis and anchor pelvic floor muscles.

Spinal Movements & Load Mechanics



Each spinal region has a distinct movement profile shaped by facet joint orientation and disc geometry.

> **Key Biomechanical Insight:** Intervertebral discs absorb up to **70% of axial compressive load**. Poor posture multiplies disc stress — forward head posture can increase cervical load by 10×.

Flexion

Anterior disc compression; posterior ligament tension

Extension

Posterior disc compression; facet joint loading

Lateral Bending

Asymmetric disc and facet loading

Key Takeaways

Joint Architecture

Synovial joints dominate human movement, relying on cartilage, fluid, and capsule for smooth, low-friction motion.

Force & Stress

Compressive, tensile, shear, and torsional forces act simultaneously on joints — understanding them is key to injury prevention.

Elbow Mechanics

A hinge joint with complex rotation; overuse injuries like epicondylitis are common in repetitive activities.

Shoulder Mobility

The most mobile joint relies on the rotator cuff for dynamic stability — a trade-off that creates vulnerability.

Spinal Column

Regional specialization — cervical mobility, thoracic stability, lumbar load-bearing — defines spinal biomechanics.